

Project 2: Full-Stack Application

Start Assignment

- Due Mar 11 by 2:59am
- Points 200
- Submitting a text entry box, a website url, or a file upload

Project 2: Full-Stack Application

Weight: 20% of final grade | **Points:** 200

Objective

Integrate multiple AI modalities with professional engineering practices to build a complete full-stack application.

Approval Requirement

Project idea must be approved by the professor on the #projects Slack channel at least one week before the deadline.

Requirements

Functional Requirements

- Complete full-stack (frontend + backend + database)
- User authentication (JWT or OAuth)
- 3+ distinct features/roles
- Real-time updates OR complex state management
- Public API with documentation
- Professional UI/UX

Technical Requirements

- Use all 3 AI modalities appropriately
- Tech Stack: React/Next.js + TailwindCSS, Node.js/Express or Next.js API routes, PostgreSQL or MongoDB
- 80%+ test coverage (Unit + Integration + E2E)
- Evaluation suite with automated tests, code quality metrics, security scanning
- Advanced CI/CD: multi-stage pipeline, deploy previews, coverage reporting

Agile Process

- 2+ documented sprints

- Sprint planning documents and retrospectives
- User stories with acceptance criteria

Deliverables

1. GitHub repository
2. Deployed app (production URL)
3. Eval dashboard (live or screenshots)
4. Complete documentation package
5. 10-minute demo video
6. 1500-word technical blog post

Rubric (200 points)

Category	Points	Description
Functionality	45	Features complete, authentication working, API functional
Technical Excellence	60	Code quality, architecture, test coverage (80%+)
AI Mastery	30	Effective use of all 3 modalities
CI/CD & DevOps	30	Pipeline quality, deployment, monitoring
Agile Process	20	Sprint documentation, retrospectives
Documentation	15	README, API docs, blog post quality

Project 2: Full-Stack Application Rubric

Criteria	Ratings					Pts
Functionality	45 pts Excellent - All features complete, authentication working perfectly, API functional and well-designed, polished UI/UX	36 pts Good - Most features complete, authentication works, API functional, good UI/UX	27 pts Satisfactory - Core features work, basic authentication, API works, acceptable UI/UX	18 pts Needs Improvement - Some features missing, auth issues, API problems, poor UI/UX	0 pts Unsatisfactory - Major features broken or missing	45 pts
Technical Excellence	60 pts Excellent - Outstanding architecture, 80%+ test coverage, excellent database design, strong security practices	48 pts Good - Good architecture, 70-79% coverage, good database design, adequate security	36 pts Satisfactory - Acceptable architecture, 60-69% coverage, basic database design, some security	24 pts Needs Improvement - Weak architecture, <60% coverage, poor database design, security gaps	0 pts Unsatisfactory - No clear architecture, minimal tests, major security issues	60 pts
AI Mastery	30 pts Excellent - Effective use of all 3 modalities with clear documentation of when/why each was used	24 pts Good - Good use of all 3 modalities, mostly documented	18 pts Satisfactory - Used all 3 modalities but with limited effectiveness or documentation	12 pts Needs Improvement - Missing modality or poor usage, weak documentation	0 pts Unsatisfactory - Did not demonstrate multi-modality usage	30 pts
CI/CD & DevOps	30 pts Excellent - Multi-stage pipeline, deploy previews, coverage reporting, security scanning,	24 pts Good - Working pipeline with most features, good automation	18 pts Satisfactory - Basic pipeline working, some automation	12 pts Needs Improvement - Incomplete pipeline, manual steps required	0 pts Unsatisfactory - No working CI/CD pipeline	30 pts

Criteria	Ratings					Pts
	automated deployment					
Agile Process	20 pts Excellent - 2+ well-documented sprints, clear planning docs, insightful retrospectives	16 pts Good - 2 sprints documented, good planning, useful retrospectives	12 pts Satisfactory - Basic sprint documentation, some planning evident	8 pts Needs Improvement - Incomplete sprint docs, minimal planning	0 pts Unsatisfactory - No agile process documentation	20 pts
Documentation	15 pts Excellent - Comprehensive README, complete API docs, quality blog post, excellent demo video	12 pts Good - Good README and API docs, solid blog post and demo	9 pts Satisfactory - Adequate documentation across all areas	6 pts Needs Improvement - Incomplete or poor quality documentation	0 pts Unsatisfactory - Missing major documentation components	15 pts
Total Points: 200						